

Nitrogen fixation rates in temperate forest soil microbial communities

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ABSTRACT

We quantified nitrogen fixation in temperate forest soils. Fixation rates increased with soil moisture across all sampled plots. Microbial community composition shifted seasonally.

METHODS

Soil cores (n = 48) were collected across four plots and assayed by acetylene reduction.

RESULTS

Nitrogen fixation rates increased significantly with soil moisture ($r = 0.62$, $p = 0.001$). Diazotroph abundance was higher in autumn samples.